Chapter 7 solutions

CH07-E1

1.

$$A^TA = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}.$$

Eigenvalues $3, 2$, eigenvectors $v_1 = (1,0)^T, v_2 = (0,1)^T$.

2.

$$\sigma_1 = \sqrt{3}, \qquad \sigma_2 = \sqrt{2}.$$
$$u_1 = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad u_2 = \frac{1}{\sqrt{2}}(1,-1,0)^T.$$

3.

$$A = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 0 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

4. Complete the basis with

$$u_3 = (1,1,-2)^T/\sqrt{6}.$$

Full column rank gives $N(A) = \{0\}$, while $\dim N(A^T) = 3 - 2 = 1$. The vector u_3 is a unit left-nullspace basis vector. The full rectangular diagonal factor is

$$\begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}.$$

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CH07-E2

1.

$$m = 3, n = 2, r = 2.$$
$$\dim C(A^T) = 2, \quad \dim N(A) = 0,$$
$$\dim C(A) = 2, \quad \dim N(A^T) = 1.$$

2. Bases:

- $C(A^T)$:e_1, e_2;
- $N(A)$ is the zero subspace, whose basis is the empty set, not a set containing zero.
- $C(A)$:u_1, u_2;
- $N(A^T)$:u_3.

3.

Chapter 7 solutions (continued)

$$Av_1 = \sqrt{3}u_1, \qquad Av_2 = \sqrt{2}u_2.$$

4. The three left vectors form an orthonormal basis of $\mathbb{R}^3$, giving the column-space and left-nullspace sum. The row space already equals $\mathbb{R}^2$, with trivial nullspace.

CH07-E3

1. Columns dependent, rank 1:

$$u_1 = \frac{1}{\sqrt{5}}(1,2)^T, \qquad v_1 = \frac{1}{\sqrt{5}}(1,2)^T.$$
$$A = 5u_1v_1^T, \qquad \sigma_1 = 5.$$

2.
$$5u_1v_1^T = A.$$

3.
$$\|A\|_2 = 5, \qquad \|A\|_F = 5.$$
$$A^+ = \frac{1}{5}v_1u_1^T = \frac{1}{25}\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}.$$

4.
$$x^+ = A^+b = \frac{7}{25}(1,2)^T.$$

Normal equation:

$$x_1 + 2x_2 = \frac{7}{5}.$$

All minimizers are

$$x = x^+ + t(-2,1)^T, \quad t \in \mathbb{R}.$$

The added vector lies in the nullspace and is orthogonal to x^+. Pythagoras proves that x^+ uniquely minimizes the norm.

CH07-E4

1. Every finite real rectangular matrix has an SVD. Here $|\det A| = 20$ equals the product of its two singular values, so neither is zero and A is invertible.
2. For a full-rank square matrix,

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_2} \geq 1.$$

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Chapter 7 solutions (continued)

3. A positive singular value becoming zero makes the matrix rank deficient and singular. The ordinary inverse condition number becomes infinite. Changes in the other singular values depend on the perturbation.

CH07-E5

1. Three singular values are positive, so the rank is 3.
2. Spectral errors:

$$k = 1 : 4, \qquad k = 2 : 1, \qquad k = 3 : 0.$$

Frobenius errors:

$$k = 1 : \sqrt{17}, \qquad k = 2 : 1, \qquad k = 3 : 0.$$

3. Retained energy:

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_r^2}.$$

4. The spectral error is σ_{k+1}. Since $\sigma_2 = 4$, rank 1 suffices, while rank 0 has error 9. Thus the minimum is $k = 1$.

CH07-E6

1. Each row sums to zero, so the samples are centered.
2.

$$X_0 X_0^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

$$u_1 = (1, 1)^T/\sqrt{2}, \qquad u_2 = (1, -1)^T/\sqrt{2}.$$

Scores:

$$s_1 = (-\sqrt{2}, 1/\sqrt{2}, 1/\sqrt{2}),$$
$$s_2 = (0, -1/\sqrt{2}, 1/\sqrt{2}).$$

3.

$$\widehat{C} = \frac{1}{2}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The first and second component sample variances are $3/2$ and $1/2$, respectively.

4. Scaling row 2 by 100 makes its variance dominate, moving the first direction toward the second coordinate axis. Standardization avoids domination by units but replaces covariance geometry by correlation geometry. That choice depends on the application.

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Chapter 7 solutions (continued)

CH07-E7

1.
$$A = \sqrt{10}\, u_1 v_1^T, \quad u_1 = (1,2)^T/\sqrt{5}, \quad v_1 = (1,1)^T/\sqrt{2}.$$

2.
$$A^+ = \frac{1}{\sqrt{10}} v_1 u_1^T = \frac{1}{10}\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}.$$
$$x^+ = A^+ b = \frac{7}{10}(1,1)^T.$$
$$AA^+ b = \frac{7}{5}(1,2)^T.$$

3. Residual:
$$e = \left(-\frac{2}{5}, \frac{1}{5}\right)^T.$$
$$e^T(1,2) = 0.$$

4. All minimizers:
$$x = x^+ + t(-1,1)^T, \quad t \in \mathbb{R}.$$

The minimum-norm solution lies in row space, perpendicular to nullspace. Adding a nonzero nullspace vector increases its squared norm.

CH07-E8

1. Insert $V^T V = I$ into the SVD.
$$A = U\Sigma V^T = (UV^T)(V\Sigma V^T) = QS.$$

2. A product of orthogonal square matrices is orthogonal.
$$Q^T Q = I.$$
$$S^T = V\Sigma^T V^T = S.$$

For any x,
$$x^T S x = \sum_i \sigma_i (V^T x)_i^2 \geq 0.$$

3. Invertibility makes every singular value positive, proving positive definiteness and invertibility of S.

4.
$$\det A = \det Q \det S.$$

Chapter 7 solutions (continued)

Since $\det S > 0$ in the invertible case, $\operatorname{sgn}(\det A) = \operatorname{sgn}(\det Q)$. A negative determinant of Q records orientation reversal.

CH07-E9

1.

$$AA^{+} = U_r U_r^T, \qquad A^{+}A = V_r V_r^T.$$

Both matrices are symmetric, and

$$(U_r U_r^T)^2 = U_r U_r^T,$$

The same calculation applies to $V_r V_r^T$. Hence both are orthogonal projectors.

2.

$$AA^{+}A = U_r U_r^T A = A.$$
$$A^{+}AA^{+} = V_r V_r^T A^{+} = A^{+}.$$

3. If the cutoff splits a tie, different retained directions within the tied subspace give the same optimal error. The complete tied subspace remains fixed, but individual axes and the retained part of it need not be unique.
4. Flipping only u_1 changes the first term from

$$\sigma_1 u_1 v_1^T$$

to its negative. Flipping v_1 as well supplies a second minus sign and preserves the factorization.

CH07-E10

1. Eckart–Young gives $\sigma_i(A) \le \|A - C\|_2$ for every C of rank less than i. Choose $C = B_{i-1}$. Then $\sigma_i(A) \le \|A - B\|_2 + \|B - B_{i-1}\|_2 = \|A - B\|_2 + \sigma_i(B)$. Exchanging A, B and combining the inequalities proves the bound.
2. The first two singular values exceed 10^{-10}, so every compatible B has rank at least 2. The rank-2 model $B_2 = \operatorname{diag}(1, 10^{-8}, 0)$ has distance 10^{-12}; the rank-3 model $B_3 = A$ has distance zero. Both are compatible, so the third direction is unresolved. A truncated pseudoinverse retaining the second direction still amplifies it by 10^8; retention also requires an application error budget.
3. $\sigma_1 \ge \|Re_2\|_2 = \sqrt{10^{12} + 1} > 10^6$, and $\sigma_1\sigma_2 = |\det R| = 1$, so $\sigma_2 < 10^{-6}$. Without computing an SVD, an explicit rank-1 model is

$$B = \begin{bmatrix} 1 & 10^6 \\ 10^{-6} & 1 \end{bmatrix} = \begin{bmatrix} 10^6 \\ 1 \end{bmatrix} \begin{bmatrix} 10^{-6} & 1 \end{bmatrix}.$$

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Chapter 7 solutions (continued)

Only the lower-left entry changes, so $\|R - B\|_2 = 10^{-6}$. Since R is already upper triangular, ordinary QR permits $Q = I$. Its diagonal entries are both 1 despite its proximity to deficient rank. QR diagonals cannot replace singular values.

4. If $\sigma_i(A) > t + \delta$, then $\sigma_i(B) \geq \sigma_i(A) - \delta > t$. If $\sigma_i(A) < t - \delta$, then $\sigma_i(B) \leq \sigma_i(A) + \delta < t$. Within the band, perturbations may cross the threshold; equality depends on the strict or nonstrict convention. If the threshold itself changes with matrix scale, include that change as well.

CH07-E11

Successive coordinates are (1,0), (4/5,−3/5), (12/5,−6/5), (12/5,6/5). The final multiplication gives ((36+24)/25, (48−18)/25)=(12/5,6/5). Directly, $A = \frac{1}{5}\begin{bmatrix} 12 & -1 \\ 6 & 12 \end{bmatrix}$, whose first column gives the same answer. Norms are 1,1,6/$\sqrt{}$5,6/$\sqrt{}$5: both orthogonal factors preserve length, while Σ changes it.

For arbitrary unit x, put w=V^{T}x. Then $w_1^2+w_2^2=1$ and $\|Ax\|^2 = 9w_1^2 + 4w_2^2 = 4 + 5w_1^2$. Hence $2 \leq \|\,Ax\,\| \leq 3$. Equality at 3 requires $w_2=0$ and $w_1=\pm 1$, equivalent to $x=\pm v_1$; equality at 2 requires $w_1=0$ and $w_2=\pm 1$, equivalent to $x=\pm v_2$. Here both w components are nonzero, so the output length lies strictly between the two singular values. Singular values are the lengths attained along right singular directions, not a list of all possible output lengths.

CH07-E12

1. Unit-input error versus a squared sum

$A_1 = \mathrm{diag}(8,0,0)$, $E = \mathrm{diag}(0,3,2)$. For every unit x,

$$\|Ex\|^2 = 9x_2^2 + 4x_3^2 \leq 9(x_1^2 + x_2^2 + x_3^2) = 9.$$

Input e_2 attains equality, so $\|E\|_2 = 3$. Meanwhile $\|E\|_F^2 = 0 + 9 + 4 = 13$. Input e_2+e_3 has length $\sqrt{}2$, not one. Normalization gives output length $\sqrt{13/2} < 3$, consistent with the spectral bound.

2. A strict gap does not force spectral uniqueness

$$\|A - B_c\|_2 = \max\{|8-c|, 3, 2\}, \qquad \|A - B_c\|_F^2 = (8-c)^2 + 13.$$

The global spectral lower bound is three. This family attains it exactly when $|8 - c| \leq 3$, or $5 \leq c \leq 11$. Every such c is nonzero, so B has rank one. The global Frobenius bound $\sqrt{}13$ is attained within this family only at c=8. We have classified only diagonal candidates Bc, not excluded other spectral minimizers.

Chapter 7 solutions (continued)

3. A kernel witness for a nondiagonal candidate

Take $z = (1, -1, 0)^T/\sqrt{2}$. Direct multiplication gives Bz=0 and $Az = (8, -3, 0)^T/\sqrt{2}$, hence

$$\|(A - B)z\|^2 = \|Az\|^2 = 73/2 \geq 9.$$

Since z is unit, $\|A - B\|_2 \geq \sqrt{73/2} > 3$: this B is not optimal. A single unit kernel witness suffices for the lower-bound argument; B's SVD is unnecessary.

4. Global uniqueness in Frobenius norm

Let P project onto C(B), with rank at most one. Set $\alpha_i = \|Pe_i\|^2$. Then $0 \leq \alpha_i \leq 1$ and their sum is at most one. Thus

$$\operatorname{tr}(PAA^T) = 64\alpha_1 + 9\alpha_2 + 4\alpha_3 \leq 64\alpha_1 + 9(1 - \alpha_1) \leq 64.$$

Attaining squared Frobenius error 13 requires trace 64. This forces α_1=1 and then α_2=α_3=0. Therefore P projects onto span(e_1). Columnwise Pythagoras,

$$\|A - B\|_F^2 = \|(I - P)A\|_F^2 + \|PA - B\|_F^2,$$

also forces its second term to zero. Hence B=PA=diag(8,0,0), uniquely among all allowed matrices, not merely among the diagonal family.

5. A cutoff through a repeated singular value

Let $w_\theta^\perp = (-\sin\theta, \cos\theta, 0)^T$. In the orthonormal basis $w_\theta, w_\theta^\perp, e_3$, D acts by 3,3,1 and Cθ by 3,0,0. The error therefore acts by 0,3,1:

$$\|D - C_\theta\|_2 = 3, \qquad \|D - C_\theta\|_F = \sqrt{3^2 + 1^2} = \sqrt{10}.$$

Both attain the lower bounds. Angles zero and π/2 retain different axes and produce different matrices. Angles θ and θ+π give the same projector, so not every angle produces a distinct matrix. The complete singular subspace for value three is always span(e_1,e_2); its selected one-dimensional part changes.

Chapter 7 solutions (continued)

Verification code appendix: svd_and_pca.py

```
"""Recompute simple SVD, low-rank PCA and pseudoinverse checks."""

from math import sqrt


Vec = tuple[float, ...]
Mat = tuple[tuple[float, ...], ...]


def transpose(a: Mat) -> Mat:
    return tuple(zip(*a))


def matmul(a: Mat, b: Mat) -> Mat:
    return tuple(
        tuple(
            sum(a[i][k] * b[k][j] for k in range(len(b)))
            for j in range(len(b[0]))
        )
        for i in range(len(a))
    )


def fmt(value: float) -> float:
    return 0.0 if abs(value) < 1e-12 else round(value, 12)


def clean(a: Mat) -> Mat:
    return tuple(tuple(fmt(x) for x in row) for row in a)


a = ((1.0, 1.0), (1.0, -1.0), (1.0, 0.0))
u = (
    (1 / sqrt(3), 1 / sqrt(2)),
    (1 / sqrt(3), -1 / sqrt(2)),
    (1 / sqrt(3), 0.0),
)
sigma = ((sqrt(3), 0.0), (0.0, sqrt(2)))
v = ((1.0, 0.0), (0.0, 1.0))
reconstructed = matmul(matmul(u, sigma), transpose(v))
assert clean(reconstructed) == a

rank_one = ((1.0, 2.0), (2.0, 4.0))
```

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Chapter 7 solutions (continued)

```
u1 = (1 / sqrt(5), 2 / sqrt(5))
v1 = (1 / sqrt(5), 2 / sqrt(5))
rebuilt_rank_one = tuple(
    tuple(5.0 * u1[i] * v1[j] for j in range(2))
    for i in range(2)
)
assert clean(rebuilt_rank_one) == rank_one

centered = ((-1.0, 0.0, 1.0), (-1.0, 1.0, 0.0))
pc1 = (1 / sqrt(2), 1 / sqrt(2))
pc2 = (1 / sqrt(2), -1 / sqrt(2))
scores1 = tuple(
    sum(pc1[i] * centered[i][j] for i in range(2))
    for j in range(3)
)
scores2 = tuple(
    sum(pc2[i] * centered[i][j] for i in range(2))
    for j in range(3)
)
assert abs(sum(x * x for x in scores1) - 3.0) < 1e-12
assert abs(sum(x * x for x in scores2) - 1.0) < 1e-12

b = (1.0, 3.0)
a_plus_times_b = (7.0 / 25.0, 14.0 / 25.0)
projection = tuple((7.0 / (5.0**0.5)) * u1[i] for i in range(2))
assert clean((tuple(a_plus_times_b),))[0] == a_plus_times_b
assert clean((projection,))[0] == (7.0 / 5.0, 14.0 / 5.0)

exercise7 = ((1.0, 1.0), (2.0, 2.0))
exercise7_plus = ((0.1, 0.2), (0.1, 0.2))
exercise7_x = matmul(exercise7_plus, ((1.0,), (3.0,)))
assert clean(exercise7_x) == ((0.7,), (0.7,))
assert clean(matmul(exercise7, exercise7_x)) == ((1.4,), (2.8,))
assert clean(matmul(matmul(exercise7, exercise7_plus), exercise7)) == exercise7
u3 = (1 / sqrt(6), 1 / sqrt(6), -2 / sqrt(6))
assert all(abs(sum(a[i][j] * u3[i] for i in range(3))) < 1e-12 for j in range(2))

print("reduced SVD reconstruction =", reconstructed)
print("PCA scores =", scores1, scores2)
print("A plus b =", a_plus_times_b)
print("projection =", projection)
```

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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